LUCAS SETO

248-832-2708 | lucas.seto@uwaterloo.ca

TECHNICAL SKILLS

Languages:	Java SE 23, Python 3.13, C++, Processing 4.4, LaTex
CAD & 3D:	2D Drafting (AutoCAD 5+ years), 3D Modelling (AutoCAD, SOLIDWORKS, Blender, Lumion)
CAM:	CNC Machining (Fusion360)
3D Printing:	Bambu Studio, H2S, PIS, A1 Hardware Frameworks
Dev Tools:	GitHub, Visual Studio Code, Arduino IDE
Microsoft Office:	Excel (Intermediate), Word, PowerPoint, Outlook
Circuitry:	Point-to-Point Soldering, Through Hole Technology (THT), Component Placement
Lab:	Micro Pipetting / Autopipetting

EDUCATION

BASc, Nanotechnology Engineering | University of Waterloo **Sep 2026 – May 3031**

OSSD | Vincent Massey Secondary School, Windsor, ON **Sep 2021 – Jun 2026**

EXPERIENCE

Formula Nano Design Team | University of Waterloo **Sep 2026 – Nov 2026**

STM (Scanning Tunneling Microscope) Construction and Electronics

- Integrated piezoelectric control device using conductive adhesives for movement on the nanometer length scale.
- Led the wiring, soldering and integration of the main PCB and auxiliary circuit boards.
- Interpreted complex electrical schematics to guide circuit assembly, component connections, and system integration.

Computational Chemistry and Molecular Modelling

- Utilized Avogadro2 to generate and export molecular coordinate files for quantum chemistry simulations using ORCA.
- Employed ORCA quantum chemistry software to optimize molecular geometries, preparing molecular structures for subsequent computational simulations.
- Executed Nudged Elastic Band (NEB) transition-state (TS) calculations in ORCA to characterize reaction pathways and generate visualizations of transition-state configurations.

STEAM Innovation Challenge Competitor | University of Toronto **Jan 2026 – May 2026**

- Designed an innovative public transit vehicle concept leveraging contemporary electric Vertical Takeoff and Landing (eVTOL) systems research.
- Constructed miniature metropolis to physically demonstrate vehicular interactions and system integration.
- Co-authored a technical research report detailing core analytical parameters and advanced air mobility data sets.
- Presented the technical data to an evaluation panel consisting of industry experts and post-doctoral researchers.

Assistant Architect | Internship at MMA Architect Inc. (AutoCAD) **Feb 2025 – Jun 2025**

- Drafted complete architectural drawing packages encompassing Floor, Foundation, Site, and Elevation plans.
- Conducted on-site structural surveying to establish accurate baseline parameters for digital drafting.
- Collaborated with licensed architects to execute project pipelines and align blueprints with community guidelines.
- Liaised with municipal authorities across Windsor, LaSalle, and Wallaceburg to navigate zoning bylaws and verify regulatory compliance.
- Managed and Organized company wide digital storage to enable quick and easy access to previous project documents.

Massey Hackathon Participant | Java, Python, Git, VS Code **May 2024, May 2026**

- Programmed a functional graphing calculator application integrating algebraic and calculus-based computations.
- Cooperated within an agile developer group to successfully build, test, and package an interactive video game.
- Applied Object-Oriented Programming paradigms using graphical libraries, including Java Swing, Pygame, and Tkinter.

VOLUNTEERING

Tournament Arbiter | University of Windsor Chess Tournament **Jun 2023**

- Supervised fair-play regulations and oversaw competitive tournament matches for a multi-round chess event.
- Resolved competitive ruling disagreements and participant conflicts with impartiality and clear administrative authority.
- Coordinated auditorium infrastructure setups, structural seating arrangements, and post-event inventory cleanups.

PROFESSIONAL CERTIFICATIONS

- Online WHMIS 2015 Review Course Certification **Completed 2026**
- Ontario Online Basic Workplace Health & Safety Certification **Completed 2023**